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BE (6th Semester)

Examination, April-May, 2016

(Old Scheme)

Advanced Electronic Circuits

Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Solve any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

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1. (*a*) Define sample period and holding period related to sample and hold circuits. 2

(*b*) Explain in detail R-2R ladder network type digital to analog converter and derive expression for analog output signal. 7

(*c*) Explain voltage to frequency converter. 7

(*d*) Explain the operation of dual slope ADC. 7

(Turn Over)

2. (*a*) What do you mean by bilinear transfer function? 2

(*b*) Draw four op-amp biquad circuits and find its frequency response. 7

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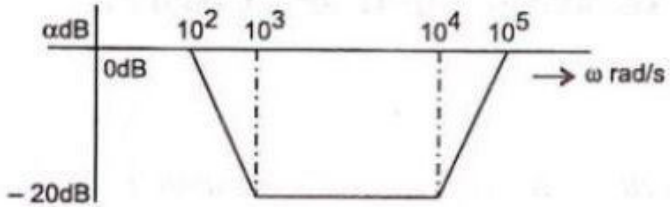
(*c*) Design a band-pass filter whose transfer characteristic satisfies the plot as shown in the diagram. Assume the capacitor value as 0.01 μF and attenuation is provided in the range 1000 rad/s ≤ ω ≤ 10000 rad/s. 7



(*d*) Design a circuit to realize the bilinear transfer function with a zero at f_z = 830 Hz, a pole at f_p = 13 kHz and a high frequency gain of 23 dB. Scale the frequency parameter by 2000 πrad/s, capacitor C_1 as 50 nF. 7

3. (*a*) What is RC-CR transformation? 2

(*b*) Design a Delyiannis-Friend band pass circuit with f_0=12.5 kHz, Q_0=10 and midband gain H=26 dB. Assume gain as 19.95 and C=5nF. 7

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(*c*) Find sensitivity of the Sallon-Key Circuit. 7

(*d*) Explain Notch filter. What are its advantages? 7

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4. (*a*) Define lock range and capture range. 2

(*b*) Describe any three applications of PLL. 7

(*c*) Explain Voltage Controlled Oscillator. 7

(d) A PLL has a VCO with K_0=25 kHz/V and f_c = 50 kHz. The amplifier gain is A=2 and the phase detector has a maximum output voltage swing of ±0.7V. Find the lock range of the PLL. Assume filter gain as unity. 7

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5. (*a*) Define multiplier and draw its equivalent circuit. 2

(*b*) Explain basic analog multiplication technique. 7

(*c*) Explain square rooting circuit. 7

(*d*) Write a short note on monolithic multiplier circuits. 7

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